

SALIENT CHARACTERISTICS

INHALED NITRIC OXIDE

The following outlines the essential characteristics for an inhaled nitric oxide (iNO) therapy system for use at the Brooke Army Medical Center. The primary purpose of this system is to act as a selective pulmonary vasodilator to improve oxygenation and blood flow in patients, particularly those with pulmonary hypertension.

List of Characteristics

- The system shall improve patient oxygenation and reduce the need for extracorporeal membrane oxygenation (ECMO).
- Be approved for treating term and near-term neonates (>34 weeks gestation) with hypoxic respiratory failure associated with pulmonary hypertension.
- The inhaled gas Shall act as a vasodilator.
- Compatibility Must be usable with additional ventilatory support therapies to maximize patient oxygen delivery.
- Shall allow for operator-determined concentration of nitric oxide, which must remain constant throughout the respiratory cycle.
- Shall be able continuously monitor inspired levels of Nitric Oxide (NO), Nitrogen Dioxide (NO₂), and Oxygen (O₂).
- Shall facilitate the monitoring of patient's partial pressure of oxygen in arterial blood (PaO₂) and methemoglobin levels during administration.
- Shall prevent the generation of inhaled Nitrogen Dioxide (NO₂) in concentrations exceeding 20 ppm.
- This shall include a backup battery power supply and a reserve nitric oxide delivery system for use in case of system or power failure.
- Shall meet all U.S. Food and Drug Administration (FDA) required specifications for this type of medical device.
- Shall be a blend of nitric oxide and nitrogen, available in concentrations such as 800 ppm (0.08% NO, 99.92% N₂) or 100 ppm (0.01% NO, 99.99% N₂), administered by inhalation.
- It shall be supplied in aluminum cylinders as a compressed gas under high pressure (2000 pounds per square inch gauge [psig]).
- Shall be available in a size 88 cylinder (or equivalent) containing at least 1963 liters at STP (delivering 1918 liters) of 800 ppm nitric oxide."